

**Analyzing the NIST Cybersecurity Framework's (CSF) Effectiveness: An Integrated
Policy Perspective**

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Introduction

The National Institute of Standards and Technology CSF has grown into a foundation of cybersecurity policies, by providing a voluntary and organized way to manage cyber risk among numerous sectors. While it was originally intended to defend critical infrastructure, its effect has spread worldwide. Still, determining its effectiveness is a challenging task. A comprehensive assessment needs to consider evaluations by cybersecurity experts, along with ethical, political, and social implications that have been observed. Ethically, the NIST CSF needs to satisfy security, as well as privacy and fairness concerns. Politically, it needs to have a decentralized governance design, even though this results in accountability concerns. Socially, its organizational impact is inconsistent, due to varying workplace culture and learning processes. This paper examines how experts have assessed the NIST CSF, finds new implications from those assessments, and suggests a way to evaluate its effectiveness.

Including Previous Policy Implications

Previous analyses of the framework have shown a variety of ethical, political, and social implications. Ethically, the NIST CSF enhances cyber awareness, but it also leads to concerns about privacy issues. Politically, the NIST CSF's voluntary style was affected by attempts to satisfy both various industries and federal standards, which has led to adoption inconsistencies. Socially, its implementation typically depends on organizational culture, employee training, and differences between various sectors. Such implications affect the assessment's standards: an effective policy needs to improve not only technological defenses, but also promote ethical conformity, political accountability, and remain socially inclusive. Realizing these separate

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characteristics leads to a more accurate evaluation that observes how the framework cooperates with organizational and human dynamics in the real world.

Scholarly Assessments of the NIST CSF

Experts provide different views on the effectiveness of the NIST CSF. Gordon, Loeb, and Zhou (2020) commend its effect, by describing how the policy “has rapidly become one of the, if not the, most widely accepted approaches to facilitate cybersecurity risk management within organizations” (p. 2). Still, they also suggest improvements, by explaining that “combining the GL [Gordon–Loeb] Model with the NIST Cybersecurity Framework could go a long way toward facilitating NIST’s recommendation that cost-benefit analysis be used” (p. 8). Fadya and Utama (2025) evaluate the NIST CSF’s capabilities within the public sector and mention noticeable performance gaps: “The current maturity level for the five functions is at Level 1 (initial) and the desired maturity level is Level 2 (managed)” (p. 191). Collectively, both studies provide evidence of the NIST CSF’s impact, and the current challenges associated with its implementation.

My Approach to Assessing the NIST CSF

An evaluation of the framework should include expert perspectives, maturity measurements, and feedback by stakeholders. By reflecting on the work produced by Bernardo, Malta, and Magalhães (2025), a multidimensional design should be considered, which would combine feedback from leadership roles, IT personnel, and other staff to observe the organizational impact of the NIST CSF. As they describe, “The inclusion of cybersecurity experts alongside distinct organizational groups... reinforces the scalability and adaptability of

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the proposed framework” (p. 7). A multilayered approach could consist of the Capability Maturity Model Integration (CMMI) method, which is known for being “a valuable tool for measuring the performance of information systems, aiming to identify maturity levels, pinpoint issues, and prioritize improvement initiatives” (Fadya & Utama, 2025, p. 184). This could be specifically applied to analyze the NIST CSF’s five functions: Identify, Protect, Detect, Respond, and Recover. Because each organization is expected to either self-assess or use external security evaluations, cybersecurity leadership teams could use the CMMI method to assess the implementation of the NIST CSF in their organization, while also considering ethical oversight, political compliance, and social responsiveness. Therefore, the framework’s adoption should not only be analyzed from a technical perspective, but also whether it facilitates positive changes in organizational decision-making processes and other employee behaviors. These ideas would help organizations assess the implementation of the current version of the NIST CSF, allowing them to provide feedback that would help NIST develop future versions of the framework.

Predicting the Outcome of this Assessment

If the NIST CSF were assessed in this manner, it would show some success, but with inconsistencies. Most evaluations support its foundational strengths, which has partially led to its widespread adoption. Nonetheless, its application in the real world has shown some gaps, particularly within low-maturity organizations with a limited amount of financial or technical resources. Fadya and Utama (2025) highlight how “the information system management team... need to have awareness, understanding, and commitment” to reach desirable maturity levels (p. 190). Bernardo et al. (2025) support these suggestions by adding that “proactive steps, such as establishing strong security policies, improving employee education, and consistently updating

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systems” are vital for the success of any cybersecurity policy (p. 17). These insights from both studies suggest that while the NIST CSF provides guidance for implementing the foundation of satisfactory cybersecurity systems, its effectiveness significantly relies on how each organization integrates it with their operational workflows and overall culture. Therefore, a multidimensional evaluation could reveal gaps in the framework’s implementation to identify improvements in its accessibility and effectiveness.

Conclusion

An assessment of this framework involves more than ensuring compliance; it also requires maintaining an ethical balance, political compliance, and its effective integration from a social standpoint. By analyzing various scholarly assessments with a multidimensional strategy, we realize an accurate representation of its strengths and weaknesses. Even though the NIST CSF is widely used, it remains challenging to implement it consistently and determine improvements. A policy as prominent as the NIST CSF requires consistent efforts to ensure its effectiveness through adaptive assessment methods that consider real-world limitations, such as those relating to human, technical, and organizational realities. This will ensure that the NIST CSF remains relevant, effective, and ready to be adapted to necessary changes in the long term.

References

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